

Dependent (Paired Samples) t Test

ITAO 20200: Foundations of Statistics

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Description

A paired-samples (or dependent) t test examines differences on scores for two related samples. For instance, in looking to see if there is a change in scores after an intervention, one could compare a pretest and posttest score for the group of people exposed to the treatment. You could also evaluate scores from different people who were matched into pairs due to similarity (e.g., spouses, twins, etc.).

Assumptions

This test uses variables measured on interval or ratio scales, and it assumes they are normally distributed as well as measured on the same scale. If they are not measured on the same scale, they should be converted to z -scores before performing the t test.

Data

For this test, two separate variable columns must be created, one for each observation. We will use the **Grades.csv** data. There are five variables in this data set. **Pretest**, **Midterm**, and **Final** all represent scores obtained on those tests. **Teacher** is a grouping variable indicating which of three different instructors taught the students. **Required** is another grouping variable indicating whether or not the class was an elective or was required. A portion of the data is shown below.

```
Grades <- read.csv("https://mendoza-stats.s3.amazonaws.com/grades.csv")
head(Grades, n = 3) # First 3 observations in the data set
```

##	Pretest	Midterm	Final	Teacher	Required
## 1	56	75	77	1	1
## 2	79	91	84	1	0
## 3	68	82	81	1	1

Data Analysis & Output

Our objective is to determine if scores have changed from the **Pretest** to **Midterm**, and from **Midterm** to **Final**.

Some Summary Statistics

First, let's obtain some basic summary statistics about our variables of interest. Individual commands can be used, as in the following:

```
summary(Grades$Pretest)
sd(Grades$Pretest)
length(Grades$Pretest)

summary(Grades$Midterm)
sd(Grades$Midterm)
length(Grades$Midterm)

summary(Grades$Final)
sd(Grades$Final)
length(Grades$Final)
```

Rather than utilize repetitive code like this, one line of code can be used for all three variables at once by using the `apply` command. This shortcut applies a function (e.g., `sd`, `length`) to each variable named in the list.

```
apply(Grades[, c("Pretest", "Midterm", "Final")], summary)
```

```
##      Pretest  Midterm   Final
## Min.    50.000 72.00000 69.000
## 1st Qu.  57.000 78.50000 78.500
## Median   61.000 81.50000 84.500
## Mean     64.125 82.20833 84.625
## 3rd Qu.  70.250 85.50000 92.000
## Max.     80.000 95.00000 100.000
```

```
apply(Grades[, c("Pretest", "Midterm", "Final")], sd)
```

```
## Pretest Midterm   Final
## 8.573531 5.999849 8.355225
```

```
apply(Grades[, c("Pretest", "Midterm", "Final")], length)
```

```
## Pretest Midterm   Final
##      24      24      24
```

Paired Samples t Test

Now, let's move on to do our paired sample t -tests. R is capable of doing either one- or two-tailed t tests using the `t.test()` function. The argument that differentiates between the type of test is `alternative`. For a two-tailed test, use `alternative = "two.sided"` (which is the default setting). For a lower-tailed hypothesis test, use `alternative = "less"` option in the function. For an upper-tailed hypothesis test, use `alternative = "greater"`.

We will like to test the following hypotheses:

$$H_0 : \mu_d = 0$$

$$H_a : \mu_d \neq 0$$

using $\alpha = 0.05$. To do so in R, use the `t.test()` function as follows:

Paired Sample t -test for Pretest and Midterm

```
t.test(Grades$Pretest, Grades$Midterm, # the variables we are testing
       alternative = "two.sided",      # the alternative hypothesis
       mu = 0,                        # the null (or test) value
       conf.level = 0.95,             # the confidence level = 1 - alpha
       paired = TRUE)                 # this indicates that it's a paired t test
```

```
##
## Paired t-test
##
## data: Grades$Pretest and Grades$Midterm
## t = -19.643, df = 23, p-value = 7.16e-16
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
##  -19.98776 -16.17890
## sample estimates:
## mean difference
##      -18.08333
```

Paired Sample t -test for Midterm and Final

```
t.test(Grades$Midterm, Grades$Final, # the variables we are testing
       alternative = "two.sided",      # the alternative hypothesis
       mu = 0,                        # the null (or test) value
       conf.level = 0.95,             # the confidence level = 1 - alpha
       paired = TRUE)                 # this indicates that it's a paired t test
```

```
##
## Paired t-test
##
## data: Grades$Midterm and Grades$Final
## t = -1.8029, df = 23, p-value = 0.08453
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
##  -5.1895936  0.3562603
## sample estimates:
## mean difference
##      -2.416667
```

The output is labeled **Paired t-test**, and it provides the obtained **t value**, **degrees of freedom**, the **p -value**, the **alternative hypothesis**, the **95% CI** of the true mean difference, and the **sample mean difference**.

Interpreting the Output

If the reported p -value is less than or equal to your alpha level ($p \leq \alpha$), then there is a statistically significant difference between the groups. This suggests that there is a very low probability of obtaining such an extreme value for t if, in fact, the null hypothesis is true. If the reported p -value is greater than your alpha level ($p > \alpha$), then your results are inconclusive.

The confidence interval for the difference can also be used to determine if a statistically significant result exists. Evaluate the interval with respect to **zero** (the null value), which would indicate no difference between the two related samples. If zero falls outside of the interval, then there is a statistically significant difference between them. If zero falls inside of the interval, then there is no statistically significant difference between them.

Writing Your Results

To convey your results properly, you need to know the t value, the degrees of freedom, and the p -value of the test statistic. Additionally, it would be helpful to have the mean and standard deviation of the groups. In the example above, we could report one significant and one non-significant result.

Dependent t tests were performed to compare scores on the *Pretest* and the *Midterm* as well as the *Midterm* and the *Final*. Results indicate that scores on the *Pretest* ($\bar{x} = 64.12$, $s = 8.57$) were significantly lower than scores on the *Midterm* ($\bar{x} = 82.21$, $s = 6.00$), $t(23) = -19.643$, $p = 0.0000$. However, scores on the *Midterm* were not significantly different from scores on the *Final* ($\bar{x} = 84.62$, $s = 8.36$), $t(23) = -1.8029$, $p = 0.0845$.

Practice On Your Own

Perform a paired samples/dependent t test at a significance level of 5% to determine if:

1. there is a significant difference between the **Pretest** and the **Final**. In other words:

$$H_0 : \mu_d = 0$$

$$H_a : \mu_d \neq 0$$

2. the **Final** scores were significantly higher than the **Midterm** scores. Assume that **Final** is population 1 and **Midterm** is population 2. As such, if we are expecting **Final** scores to be higher, then our hypothesized mean difference should be a positive value. Thus, our hypotheses are:

$$H_0 : \mu_d \leq 0$$

$$H_a : \mu_d > 0$$

3. the difference between **Midterm** and **Pretest** scores was significantly less than 15 points. Assume that **Midterm** is population 1 and **Pretest** is population 2. If that is the case, then our null and alternative hypotheses would be the following:

$$H_0 : \mu_d \geq 15$$

$$H_a : \mu_d < 15$$